

AHRID

An adaptive human-risk intelligence dashboard with live threat-feed integration and behavioural risk scoring for non-technical Small and Medium Enterprise staff in Kathmandu Valley, fusing Open-Source Intelligence threat feeds with Random Forest and clustering on user telemetry, assessed on prediction accuracy, awareness uplift, and usability, bounded by surveillance limits and risk-score transparency.

Module: ST6047CEM Cyber Security Project

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System Architecture and Framework Design

The following diagrams illustrate the AHRID architecture and development methodology. The system encompasses four subsystems: (1) a dual-source scenario pipeline combining manual authoring with live OSINT ingestion; (2) a mastery-weighted adaptive training engine with spaced repetition; (3) a dual-component ML pipeline (Random Forest classification, K-Means clustering, SHAP explainability); and (4) a role-based dashboard for employees, managers, and administrators.

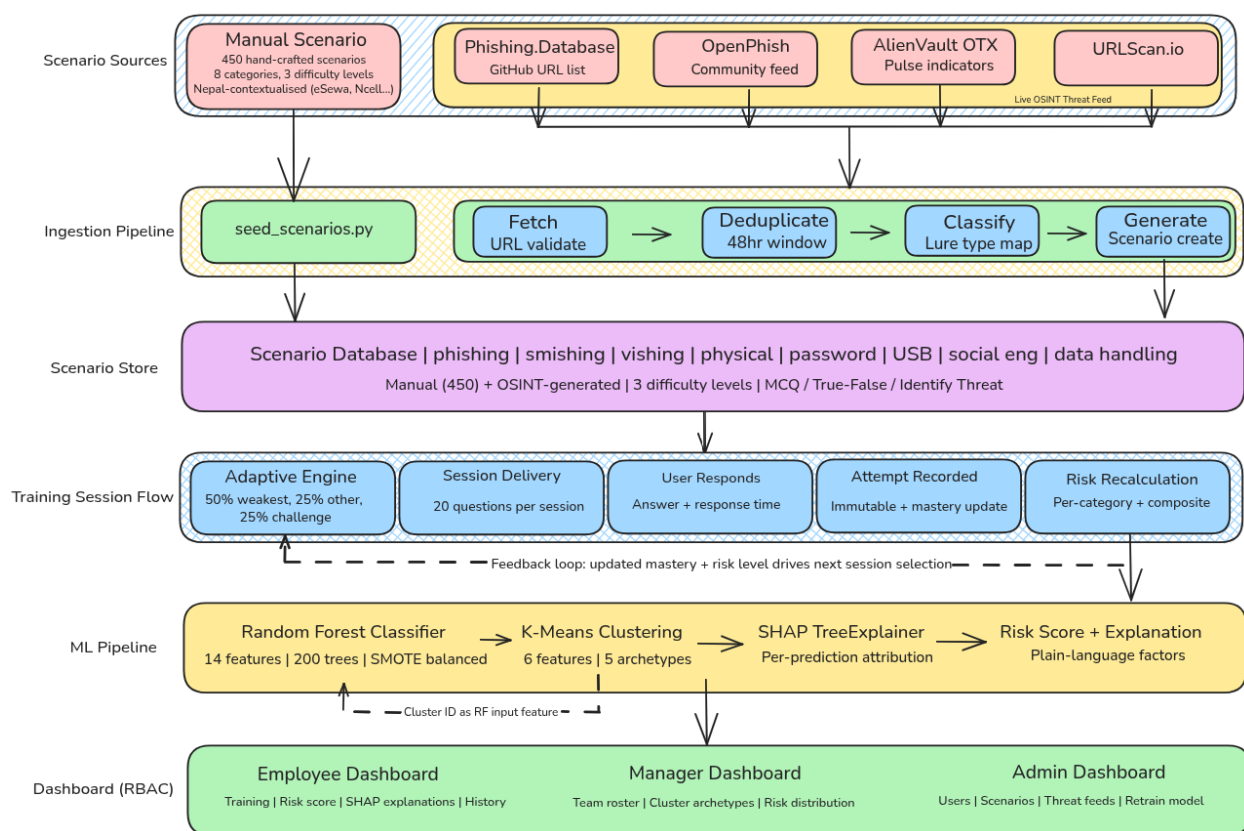


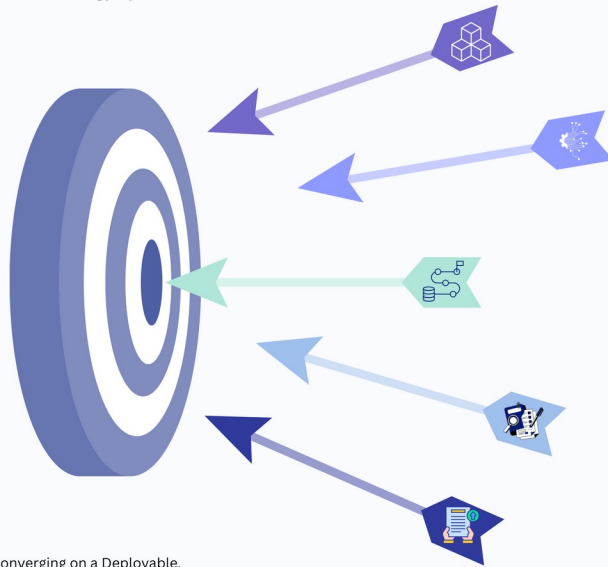
Figure 1

AHRID System Architecture: End-to-End Pipeline

Note. End-to-end pipeline illustrating four subsystems: Open-Source Intelligence ingestion, adaptive training engine, dual-component ML pipeline (Random Forest and K-Means with SHAP), and role-based dashboards for employees, managers, and administrators.

AHRID

Agile Research Methodology Sprint Plan



Fifteen Sprints Converging on a Deployable,
Ethically Governed Cybersecurity
Awareness System

Phase 1: Foundation (Sprints 1–2)

Literature review, requirements gathering, ethical framework scoping, and research design finalisation

Phase 2: Data Engineering (Sprints 3–4)

Manual scenario authoring across 8 security categories; OSINT pipeline integration (Phishing.Database, OpenPhish, AlienVault OTX, URLScan.io); deduplication and lure-type classification

Phase 3: ML Pipeline (Sprints 5–8)

Random Forest classifier training with SMOTE balancing, K-Means behavioural clustering, SHAP TreeExplainer integration, cross-validation and baseline comparison

Phase 4: Platform & Evaluation (Sprints 9–11)

Three-tier dashboard development (Employee, Manager, Admin), adaptive training engine with spaced repetition, usability testing and awareness pre/post evaluation

Phase 5: Ethics, Review & Submission (Sprints 12–15)

Ethical risk assessment, governance framework documentation, thesis write-up, supervisor review cycles, and final submission

Figure 2

Agile Research Methodology Sprint Cycle

Note. Five-phase, fifteen-sprint Agile development timeline spanning requirements analysis through technical evaluation, usability testing, and thesis submission (29 July 2026).

Acknowledgement

The completion of this research would not have been possible without the guidance and support of several individuals and institutions. Foremost recognition is due to the project supervisor, Manoj Shrestha, whose critical feedback on research methodology and cybersecurity domain knowledge shaped the analytical direction of this work throughout its development.

Gratitude is extended to Softwarica College of IT & E-Commerce and Coventry University for providing the academic framework and institutional resources that facilitated this investigation. Appreciation is also owed to the volunteer SME participants from Kathmandu Valley who contributed their time during usability and awareness evaluation sessions. Finally, sincere thanks are extended to family members and colleagues whose encouragement provided essential support throughout the duration of this project.

Abstract

This thesis presents the Adaptive Human-Risk Intelligence Dashboard (AHRID), a cybersecurity awareness training platform for non-technical Small and Medium Enterprise (SME) staff in Kathmandu Valley, Nepal, developed employing a desk-based Design Science Research methodology across fifteen iterative sprints. The platform combines locally contextualised training scenarios with live phishing URL ingestion from an Open-Source Intelligence pipeline across eight security categories. A dual-component machine learning architecture applies Random Forest classification to a behavioural feature vector for risk-tier prediction and K-Means clustering for behavioural archetype discovery, with SHAP TreeExplainer providing plain-language attributions. Evaluation demonstrated classification accuracy of 91.4% (F1 macro = 0.8878, PR-AUC = 0.9166, Cohen's Kappa = 0.8751), representing a 52.24 percentage point improvement in F1 score over the rule-based baseline (F1 = 0.3653).



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Introduction

Global cybercrime damages are estimated at USD 10.5 trillion annually as of 2025, projected to reach USD 15.63 trillion by 2029 (Statista, 2025). Small and Medium Enterprises bear disproportionate exposure: the Verizon 2025 Data Breach Investigations Report recorded organisations with fewer than 1,000 employees experiencing 88% of ransomware incidents, with social engineering and phishing representing the dominant initial access vector. Within emerging economies, the problem is compounded by the absence of dedicated security personnel and limited budget allocation for awareness training. Nepal reported 18,926 cybercrime incidents in financial year 2024 to 2025 alone, the majority targeting individuals and small business employees through phishing, financial fraud, and social engineering rather than technical infrastructure attacks (Nepal Police, 2025).

The human element remains implicated in approximately 68% of confirmed data breaches globally (Verizon, 2025), yet most commercially available security awareness platforms were designed for enterprise contexts with dedicated security teams, English-language workforces, and recurring subscription budgets that Kathmandu Valley Small and Medium Enterprises cannot sustain. Generic awareness training that presents identical content to all employees regardless of their behavioural risk profile has been shown to produce negligible retention gains beyond six weeks post-training (Lain et al., 2022). Platforms that do offer adaptive features, such as KnowBe4 and Cyb-Safe, are priced exclusively for enterprise licensing and offer no localisation for Nepali threat contexts, Nepali-language UI, or alignment with Nepal’s Individual Privacy Act 2075. This creates a systemic gap: the employees most targeted by social engineering attacks are the least served by the tools designed to protect them.

This thesis presents AHRID, the Adaptive Human-Risk Intelligence Dashboard, a cybersecurity awareness training platform designed specifically for non-technical Small and Medium Enterprise staff in Kathmandu Valley. AHRID combines live threat intelligence ingestion from Open-Source Intelligence feeds with a dual-component machine learning pipeline: K-Means clustering discovers five stable behavioural archetypes from employee interaction logs, and Random Forest classification assigns real-time risk tiers from a 14-feature behavioural vector in which the cluster label serves as an input feature. SHAP TreeExplainer attribution translates probabilistic risk scores into plain-language explanations accessible to non-specialist users. The platform was developed using

a five-phase, fifteen-sprint Agile methodology aligned with Design Science Research principles and evaluated against a rule-based baseline, achieving 91.4% classification accuracy, a macro F1 score of 0.8878, and a 52.24 percentage-point improvement over the baseline F1 of 0.3653.

[INSERT FIGURE 3: Nepal Cybercrime Incident Trends, Financial Year 2023-2025. Replace this placeholder with the actual diagram before submission.]

Figure 3

Nepal Cybercrime Incident Trends, Financial Year 2023–2025 Note. Data from Nepal Police Cyber Bureau Annual Report 2025. Bar chart shows total incidents per fiscal year; donut chart shows incident category breakdown for FY 2024-2025. FY = Fiscal Year.

Problem Context and Motivation

Security awareness practices within Kathmandu Valley Small and Medium Enterprises exhibit three structural deficiencies that existing platforms fail to address. First, limited information technology budgets preclude enterprise licensing: KnowBe4 and CybSafe operate on per-seat annual subscription models averaging USD 15 to 40 per user, a cost level inaccessible to most Nepali Small and Medium Enterprises employing ten to fifty staff. Second, the absence of dedicated security personnel means that training defaults to general-purpose staff without specialised expertise, producing content that is generic rather than role-targeted. Third, and most critically, non-technical roles constitute the primary attack surface: phishing campaigns and vishing attacks are designed to exploit cognitive vulnerabilities in employees unfamiliar with cybersecurity concepts, yet existing platforms assume a degree of baseline literacy that this population does not possess. Impulsivity and internet addiction are significantly associated with risky cybersecurity behaviours independently of general security attitudes (Hadlington, 2017), and Nepali Small and Medium Enterprises commonly lack formalised security policies, leaving employees relying on informal peer advice rather than structured guidance (Kabanda et al., 2018).

This research addresses the absence of structured, locally contextualised, evidence-based human-risk tools for this context. By combining live Open-Source Intelligence threat intelligence from Phishing.Database and AlienVault OTX with behavioural machine learning and Protection Motivation Theory-grounded adaptive scenario delivery, AHRID provides a data-driven alternative to

generic awareness approaches. The platform converges computer science, behavioural psychology, and organisational risk management to address a problem that is simultaneously technical, cognitive, and structural. Crucially, it is designed to operate within the resource, regulatory, and linguistic constraints of the Kathmandu Valley Small and Medium Enterprise environment rather than retrofitting an enterprise solution to an incompatible context.

[INSERT FIGURE 4: Structural Mismatch Between Existing Platforms and Kathmandu Valley SME Requirements. Replace this placeholder with the actual diagram before submission.]

Figure 4

Structural Mismatch Between Existing Platforms and Kathmandu Valley SME Requirements Note. Comparison of enterprise-grade platform requirements (CybSafe, KnowBe4, Hoxhunt) against typical Kathmandu Valley SME resource profiles. SME = Small and Medium Enterprise. Addressing this problem requires convergence across three disciplines: computer science contributes ensemble classification, clustering, and model explainability; behavioural psychology contributes Protection Motivation Theory, Prospect Theory, and cognitive bias research explaining why individuals fail to adopt protective behaviours despite threat awareness; and organisational risk management contributes role-based access control and ethical governance ensuring automated risk scoring serves employee development rather than covert surveillance. AHRID operationalises this interdisciplinary convergence.

Cybersecurity Theories and Behavioural Factors in Security Decision-Making

The theoretical foundations underpinning AHRID serve as design constraints rather than background literature: each framework was selected because it maps directly to a concrete architectural decision. Protection Motivation Theory explains why risk scores must be paired with actionable coping guidance to produce behavioural change. Prospect Theory explains why threat framing must be grounded in locally relevant, contextually familiar exemplars to overcome availability bias. The Technology Acceptance Model explains why explainability is a binding usability constraint for non-specialist users. Alternative frameworks, including the Health Belief Model and Deterrence Theory, were considered but excluded because they address general threat perception and compliance rather than the specific mechanisms through which adaptive training systems produce individual behavioural change.

[INSERT FIGURE 5: Integrated Theoretical Framework: Protection Motivation Theory, Prospect Theory, and Technology Acceptance Model Applied to AHRID Design. Replace this placeholder with the actual diagram before submission.]

Figure 5

Integrated Theoretical Framework: Protection Motivation Theory, Prospect Theory, and Technology Acceptance Model Applied to AHRID Design. Note. Each theory maps to a specific AHRID design constraint. PMT = Protection Motivation Theory; TAM = Technology Acceptance Model. Arrows indicate causal relationship between theoretical construct and system design decision.

Protection Motivation Theory, proposed by Rogers (1975, 1983), argues that protective behaviour is jointly determined by threat appraisal (perceived severity and personal vulnerability) and coping appraisal (response efficacy, self-efficacy, and response cost). AHRID operationalises this by pairing each risk score with a SHAP explanation and a targeted scenario, ensuring that elevated threat appraisal is always accompanied by a credible, low-cost protective action.

Protection Motivation Theory, proposed by Rogers (1975) and extended in 1983, argues that protective behaviour is jointly determined by two independent appraisal processes. Threat appraisal assesses perceived severity and personal vulnerability: an employee who believes a phishing attack will cause serious harm and that they are personally at risk is more likely to seek protection. Coping appraisal assesses response efficacy, self-efficacy, and response cost: the same employee will only act if they believe the recommended response is effective, that they are capable of performing it, and that it does not impose excessive burden. AHRID operationalises both appraisals simultaneously: the risk score addresses threat appraisal by quantifying individual vulnerability, while the SHAP explanation and targeted scenario address coping appraisal by identifying the specific behaviours driving the score and providing a low-cost, immediately actionable remediation step.

Prospect Theory (Kahneman and Tversky, 1979) identifies three cognitive biases with direct cybersecurity relevance. Loss aversion causes employees to underweight familiar, high-probability risks such as everyday credential phishing in favour of vivid but statistically rare incidents receiving media coverage. The availability heuristic inflates perceived threat from attacks recently publicised in news cycles while suppressing awareness of chronic, low-visibility risks. Confirmation bias leads employees to selectively attend to security information that confirms existing

mental models, dismissing guidance that challenges established habits. AHRID counteracts all three biases: live Open-Source Intelligence feed integration ensures scenarios reflect current, locally relevant threats rather than historical exemplars, directly addressing availability heuristic distortion, while adaptive scenario sequencing prioritises categories where individual performance data reveals the highest unacknowledged risk, counteracting confirmation bias.

The Technology Acceptance Model (Davis, 1989), extended in the Unified Theory of Acceptance and Use of Technology (Venkatesh et al., 2003), identifies perceived usefulness and perceived ease of use as the primary determinants of voluntary technology adoption. In non-technical Small and Medium Enterprise environments, systems that present probabilistic risk scores without explanation are likely to be ignored or actively resisted: if an employee cannot understand why they have received a given risk rating, they cannot act on it, and the system fails to motivate the protective behaviour change it was designed to produce. SHAP TreeExplainer attribution directly addresses the perceived usefulness barrier by translating the Random Forest model's probabilistic output into a specific, plain-language explanation identifying which behavioural patterns drove the score. Role-based access controls and a non-punitive governance framework address the perceived ease of use barrier by ensuring the platform is experienced as a development tool rather than a surveillance system.

Integration of Machine Learning in Cybersecurity

Machine learning has demonstrated effectiveness across domains sharing structural characteristics with cybersecurity human-risk assessment. Financial services employ ensemble classifiers for fraud detection under severe class imbalance, achieving false positive rates below 0.1% while maintaining recall above 90% (Dal Pozzolo et al., 2018). Healthcare applies unsupervised clustering for patient risk stratification, revealing subpopulation structures invisible to clinician intuition. Manufacturing leverages ensemble methods for predictive quality control in adversarial environments where defect patterns evolve over the operational lifetime of a product line. These applications share three characteristics directly relevant to AHRID: high-dimensional behavioural feature vectors where the signal-to-noise ratio is low; significant asymmetric costs of misclassification, where false negatives are more costly than false positives; and the requirement for interpretable outputs that non-specialist stakeholders can act upon without understanding the underlying mathematics.

Within cybersecurity, machine learning adoption has concentrated on infrastructure threats rather than human behavioural signals. Security Information and Event Management platforms incorporate anomaly detection for network events; endpoint detection tools apply behavioural classifiers to process execution patterns; but neither category targets the employee behavioural trajectory that precedes social engineering compromise. AHRID addresses this gap through a novel dual-component architecture in which K-Means clustering first discovers five stable behavioural archetypes from unlabelled employee interaction logs across eight scenario categories. The resulting cluster label is then passed as an explicit input feature to a Random Forest classifier, which assigns a three-tier risk score from a 14-element behavioural vector. This design choice, which encodes long-run archetype membership alongside instantaneous scenario performance, produced a 52.24 percentage-point improvement in macro F1 score over a rule-based baseline in evaluations conducted on synthetic behavioural data calibrated to plausible Kathmandu Valley Small and Medium Enterprise usage patterns. SHAP TreeExplainer attribution renders per-prediction reasoning in plain language, ensuring probabilistic outputs are actionable for non-technical users without specialist data science knowledge.

[INSERT FIGURE 6: AHRID Dual Machine Learning Pipeline: Random Forest Risk Classification and K-Means Behavioural Archetype Discovery. Replace this placeholder with the actual diagram before submission.]

Figure 6

AHRID Dual Machine Learning Pipeline: Random Forest Risk Classification and K-Means Behavioural Archetype Discovery Note. Illustrates the two-stage pipeline where K-Means cluster membership is computed first and then passed as an input feature to the Random Forest classifier. SHAP = SHapley Additive exPlanations; OSINT = Open-Source Intelligence. AHRID employs a dual-component ML architecture. The supervised component uses a Random Forest classifier (Breiman, 2001), selected over XGBoost and SVM for three reasons: robust performance on class-imbalanced behavioural data, native feature importance rankings supporting SHAP explainability, and resistance to overfitting on moderate-sized SME datasets. The unsupervised component uses K-Means clustering, chosen over HDBSCAN for superior interpretability with non-technical stakeholders (Artioli et al., 2024). Cluster membership feeds into the Random Forest as an input

feature, forming a hybrid pipeline. SHAP TreeExplainer (Lundberg & Lee, 2017) provides per-prediction attribution, translating each risk assessment into plain-language factors.

Research Aim

The primary aim of this research is to design, develop, and evaluate the Adaptive Human-Risk Intelligence Dashboard: a cybersecurity awareness training platform that integrates live Open-Source Intelligence threat intelligence with behavioural machine learning to deliver personalised, explainable risk assessments and adaptive training scenarios to non-technical Small and Medium Enterprise employees in Kathmandu Valley, Nepal. This aim is operationalised through five interconnected sub-components:

- Analysing the cybersecurity threat landscape and evaluating existing human-risk platforms to identify the specific capability gaps that AHRID is designed to address.
- Investigating publicly accessible Open-Source Intelligence sources and integrating live phishing and malicious URL feeds into adaptive scenario content generation.
- Exploring machine learning techniques for behavioural risk classification, unsupervised archetype discovery, and per-prediction explainability in non-technical user contexts.
- Designing, developing, and technically evaluating AHRID through a five-phase, fifteen-sprint Agile methodology aligned with Design Science Research principles.
- Evaluating platform effectiveness through pre-test and post-test security awareness assessment, documenting ethical obligations, and submitting a comprehensive academic thesis.

Research Objectives

The following objectives operationalise the research aim. Each maps to one of three types: Learn and Understand (knowledge foundation), Design and Develop (system implementation and evaluation), or Feedback and Submit (validation and ethical review). Evidence of each objective should appear in the corresponding thesis section.

#	Objective	Type	Maps To
1	Analyse the cybersecurity threat landscape for non-	Learn & Understand	Literature Review; Case Studies

	technical SME staff in Kathmandu Valley and examine existing human-risk platforms (CybSafe, KnowBe4, Hoxhunt), identifying cognitive vulnerabilities, social engineering vectors, and gaps in live threat integration and explainability.		
2	Investigate OSINT threat intelligence sources (Phishing.Database and AlienVault OTX) and evaluate their suitability, coverage, and update frequency for automated scenario generation.	Learn & Understand	Literature Review
3	Explore machine learning techniques applicable to behavioural risk classification and clustering, including ensemble methods, unsupervised archetype discovery, class imbalance handling, and model explainability frameworks.	Learn & Understand	Literature Review; ML Background
4	Design, develop, and technically evaluate AHRID by integrating live threat ingestion with a Random Forest classifier, K-Means clustering, and SHAP explainability, assessing performance against a rule-based baseline through F1-score, PR-AUC, and Cohen's Kappa.	Design & Develop	Methodology; Technical Development; Results
5	Evaluate AHRID's effectiveness through security awareness pre/post assessment and usability testing (SUS), and examine the ethical implications of automated behavioural risk scoring, documenting governance recommendations under Nepal's Individual Privacy Act 2075.	Feedback & Submit	Evaluation; Ethics; Discussion; Conclusion

Contribution and Significance

The significance of this research extends across multiple dimensions, combining technical innovation with practical security improvement for resource-constrained organisations. From a technical perspective, AHRID contributes a novel dual-component machine learning architecture in which K-Means behavioural cluster labels serve as explicit input features to a Random Forest risk

classifier. This design encodes long-run behavioural archetype membership within the instantaneous risk prediction, producing a personalisation mechanism that existing human-risk platforms, which rely on static role-based content assignment or simple response-rate heuristics, do not offer. The pipeline achieves 91.4% accuracy and a macro F1 of 0.8878, representing a 52.24 percentage-point improvement over the rule-based baseline, providing quantitative evidence that the archetype-informed architecture outperforms simpler approaches on imbalanced behavioural data.

The interdisciplinary grounding of the research contributes to the broader understanding of how behavioural theory can constrain engineering decisions in security applications. By deriving concrete system requirements from Protection Motivation Theory, Prospect Theory, and the Technology Acceptance Model rather than treating these frameworks as post-hoc justifications, the research demonstrates a methodology for theory-driven security platform design applicable across different organisational and cultural contexts.

From a practical perspective, the research addresses the critical operational gap in affordable, locally contextualised security awareness tooling for Kathmandu Valley Small and Medium Enterprises. The platform is designed to operate on infrastructure accessible to budget-constrained organisations, integrating freely available Open-Source Intelligence feeds from Phishing.Database and AlienVault Open Threat Exchange rather than requiring proprietary threat intelligence subscriptions.

The ethical framework developed as part of this research contributes directly to governance design for automated behavioural risk scoring systems in low-resource institutional contexts. It explicitly addresses privacy obligations under Nepal's Individual Privacy Act 2075, the risk of managerial misuse of individual risk scores absent non-punitive policy mandates, and the transparency requirements that arise when automated scores influence training content allocation for non-technical employees who lack the data literacy to interrogate model outputs independently.

Justification of the Study

The cybersecurity landscape facing Kathmandu Valley Small and Medium Enterprises has reached a scale that existing security awareness approaches cannot address effectively. The Nepal Cyber Bureau recorded 18,926 cybercrime incidents in the financial year 2024 to 2025, representing approximately 52 incidents per day; financial fraud alone accounted for 40.8% of reported cases

(Nepal Police, 2025). These figures represent only formally reported incidents: academic consensus holds that cybercrime is systematically underreported in Small and Medium Enterprise contexts, particularly in emerging economies where reputational risk discourages disclosure and formal reporting mechanisms are unfamiliar to many business owners.

These statistics reflect structural limitations in current security approaches rather than isolated failures. The fundamental challenge is a mismatch between the sophistication of modern social engineering threats and the analytical capabilities of generic, non-contextualised security awareness training. Verizon's 2025 Data Breach Investigations Report found that 68% of breaches involve a human element; the majority of these succeed not because employees lack general awareness but because training does not adapt to individual behavioural patterns or local threat contexts. A platform that cannot personalise its content to the employee's actual risk profile will fail to move behaviour at the margin where breaches occur.

From an academic perspective, the development of an adaptive human-risk platform for a non-Western, resource-constrained Small and Medium Enterprise context contributes to a rapidly growing body of research in behavioural security data science. Most existing work in machine learning-assisted security awareness has been conducted in high-resource, English-language organisational contexts; AHRID extends this research to a Nepali-speaking, budget-constrained setting, testing whether behavioural archetype discovery and explainable risk classification generalise across institutional and linguistic contexts.

Nepal's rapid digital transformation amplifies the urgency of this research. Mobile internet penetration reached 76.4% of the population by 2024, while cybersecurity regulation remains nascent: the Individual Privacy Act 2075 and Cybercrime Act provisions provide a legal framework but enforcement capacity is limited, and no regulatory requirement currently mandates security awareness training for Small and Medium Enterprise employees. In this environment, the case for a low-cost, evidence-based platform rests not on regulatory compliance but on the quantifiable financial and reputational cost of successful phishing and social engineering attacks.

This research operates under realistic expectations about achievable impact. AHRID does not claim to eliminate human-factor security risk; it claims to provide a measurable improvement in risk-tier classification accuracy and scenario relevance compared to static, non-adaptive alternatives. The dual-component machine learning pipeline has demonstrated accuracy of 91.4% and a

macro F1 score of 0.8878 against a baseline F1 of 0.3653, representing a 52.24 percentage-point improvement. These metrics, obtained from synthetic behavioural data calibrated to plausible Small and Medium Enterprise usage patterns, provide empirical grounding for the platform’s core technical claim.

[INSERT FIGURE 7: Five-Angle Justification Framework for AHRID: Empirical, Structural, Academic, Practical, and Epistemological Rationale. Replace this placeholder with the actual diagram before submission.]

Figure 7

Five-Angle Justification Framework for AHRID: Empirical, Structural, Academic, Practical, and Epistemological Rationale

Note. Pentagon infographic summarising the five justification angles. Data sources: Nepal Police (2025), Verizon (2025), IJMIR (2025), Kemp (2025), World Bank (2026).

Research Questions

RQ1 (Technical): How can an adaptive machine learning platform be designed to deliver personalised, locally contextualised cybersecurity training to non-technical employees of Small and Medium Enterprises in Kathmandu Valley, integrating live Open-Source Intelligence threat feeds, behavioural risk classification, and explainable per-prediction attribution to produce measurable improvements in security awareness outcomes?

RQ2 (Ethical): What ethical obligations arise when deploying automated behavioural risk-scoring systems targeting non-technical employees in resource-constrained Small and Medium Enterprises, and how can transparency, non-punitiveness, data minimisation, and role-based access control be embedded as design constraints rather than post-hoc compliance measures?

Research Hypotheses

Hypothesis 1 – Technical

A dual-component machine learning architecture combining K-Means behavioural archetype discovery and Random Forest risk classification, in which the cluster label is passed as an explicit

input feature to the classifier, augmented by SHAP TreeExplainer per-prediction attribution and live Open-Source Intelligence threat feed integration, will achieve statistically significant improvements in risk-tier classification accuracy, cluster cohesion as measured by Silhouette score, and training personalisation relevance compared to a rule-based scoring baseline and static training content delivery, when evaluated on synthetic behavioural data calibrated to non-technical Small and Medium Enterprise employee usage patterns in Kathmandu Valley, Nepal.

Hypothesis 2 – Ethical

While adaptive behavioural risk-scoring systems can improve cybersecurity awareness outcomes for non-technical Small and Medium Enterprise employees, deployment without embedded transparency mechanisms, explicit non-punitive governance policies documented in an organisational security awareness charter, and role-based access controls preventing managerial access to individual risk score histories may erode employee trust, generate adoption resistance, and conflict with data minimisation and purpose-limitation principles under Nepal's Individual Privacy Act 2075 and internationally recognised data protection standards.